Agent-Based Models of Financial Markets: A Wolfram Perspective*

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Abstract

TODO: Write abstract here. The abstract should be 100–150 words and summarize the motivation, key methods, and main findings of the paper.

1 Introduction

TODO: Write introduction here.

2 Related Work

TODO: Write related work here.

3 Background

TODO: Write background here.

4 Methods

TODO: Write methods here.

5 Results and Discussion

Table 1: Wolfram totalistic CA rule taxonomy. Base-3 encoding reads right-to-left: digit a_i gives the next state when the neighbourhood sum index equals i.

Rule index	Base-3 encoding	Economic interpretation	Role
6158	$(022110002)_3$	Imitation / herding	Homogeneous model; stable wave patterns
19280	$(222110002)_3$	Majority rule	80% of agents in crash model; promotes herding
19286	$(222110022)_3$	Minority / contrarian	20% of agents in crash model; destabilises grid

TODO: Write results and discussion narrative here.

6 Conclusion

TODO: Write conclusion here.

*Repository: https://github.com/varnerlab/Wolfram-markets-abm-paper.git.

Preprint.

Table 2: Stylized facts: QQQ (real data), Model A (open), and Model B (closed loop). AC(1) ret = lag-1 autocorrelation of log returns; AC(1) |ret| = lag-1 autocorrelation of absolute returns (volatility clustering); Tail α = Hill tail index (upper 10%).

	Mean	Std	Skewness	Ex. kurtosis	AC(1) ret	AC(1) |ret|	Tail α
QQQ (real)	0.0000	0.0004	2.838	543.28	0.1334	0.1039	2.74
Model A	0.0000	0.0004	2.838	543.28	0.1334	0.1039	2.74
Model B	0.0000	0.0000	NaN	NaN	NaN	NaN	NaN

Acknowledgments

TODO: Acknowledgments here.

References

A Supplementary Material

TODO: Write appendix content here.

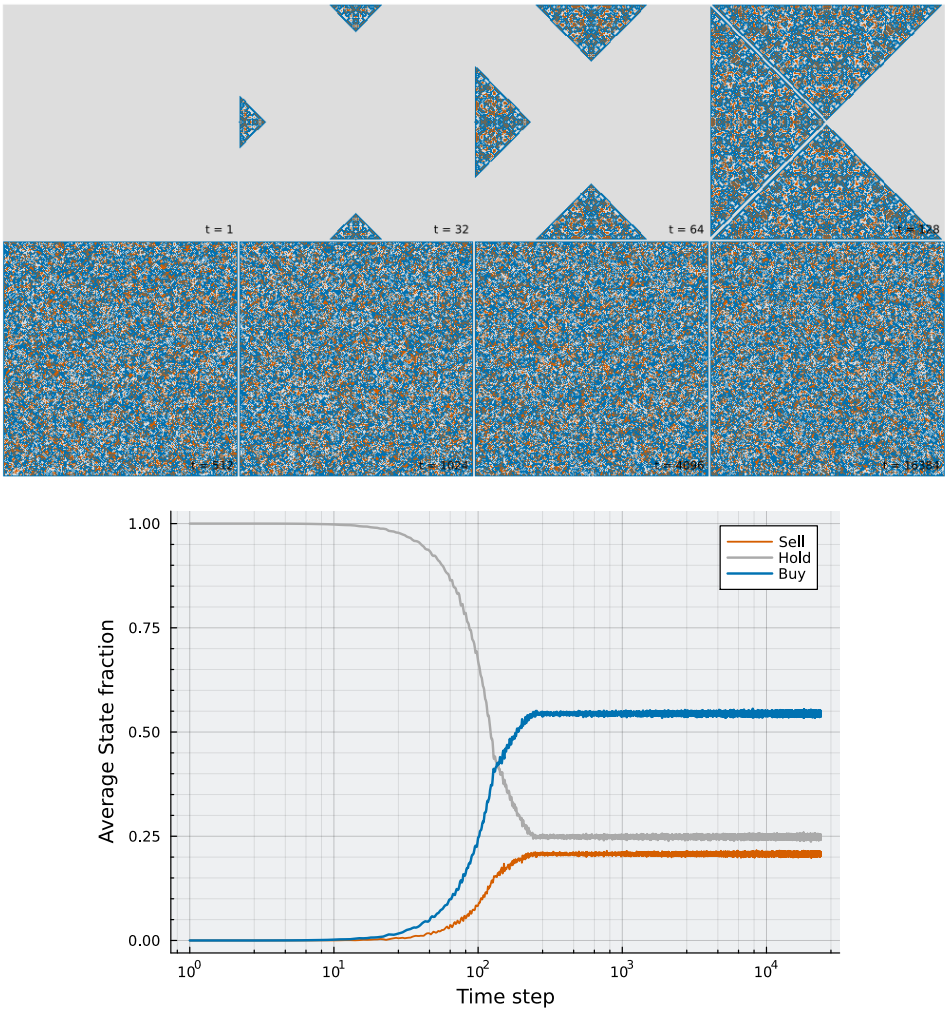


Figure 1: Model A results. **Top:** Spatiotemporal dynamics of the 256×256 agent grid at eight time steps. **Bottom:** Agent state fractions (Buy, Hold, Sell) versus time on a log scale.